

In conversation with - Pat Gelsinger

Intel's visionary CEO on the future of chipmaking

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“Obviously, we missed the mobile wave. We had like five different AI acquisitions, and when I was pushed out of the company we killed the one that would have made all the difference in the world. Those are the two monster trends that Intel missed, and we were fundamentally biased to building a great foundry. Those to me are the three massive opportunities that Intel was uniquely positioned to benefit from, and we'll do our best to bring us back into position on those fronts,” Gelsinger mentioned, when I asked him about Intel's past mistakes.

The above response is enough to tell you that Intel CEO Pat Gelsinger isn't one to mince words. After spending an hour with him, it's evident that he is the fundamental driving force behind Intel's ambitious turnaround to manufacture industry leading chips for itself and the world. After spending 30 years at Intel, from 1979 to 2009, his time away from the company also coincided with Intel's decline and decay. Ever since he returned to Intel's helm in 2021, Gelsinger's been trying to make up for lost time with the aggressive “5 nodes in 4 years”

process node strategy and beyond to restore what he calls “an iconic company” back to where it belongs.

“There's three types of semiconductor manufacturers: big, niche, or dead. And Intel's way too big to be niche, so we better be really big, if we're going to be effective for technology purposes,” Gelsinger said in response to being asked about his overall strategy.

Intel's immediate future

Intel's not only trying to make more powerful, sustainable and efficient processors and technologies under the Intel brand to power tomorrow's PCs, laptops and server data centres, but the chipmaker also wants to scale and build capacity to manufacture chips for third-party chip vendors like Qualcomm, and possibly Apple and NVIDIA as well. And the proverbial clock is ticking. Has Intel bitten off more than it can chew?

“As we go forward in time, on this Mission Impossible as it were,” Gelsinger continued with a smile. “Maybe I'm not as attractive as the Mission Impossible actor, but you know we're not hanging on the side of skyscrapers to pull this off. It's just

good, solid, disciplined engineering that's getting us to the finish line, and I think we're going to get there,” he emphasised.

Under Gelsinger's executive leadership, Intel's doing something it has never done before – overcoming significant challenges of physics to dial down several nanometer scales of semiconductor chip design and manufacturing process into a very short period of time. Intel 18A, its futuristic chip making process, is due to arrive in 2025; when Gelsinger expects the company to give the likes of TSMC and Samsung – world's two biggest contract chip makers – a tough challenge, maybe potentially overcoming them in some respect. We will have to wait and see, there's still 12-18 months to go, but Gelsinger remains confident.

After Intel delivers and meets all its aggressive timelines, “sort of climb our way back into a leadership position” as a chip manufacturer, Gelsinger believes the focus will quickly shift to “how we sustain our leadership position for years to come.” Beyond AI PC on the edge, AI accelerators in the cloud, and an innovative software stack, this will include breakthroughs in silicon wafer and chip